

QUANTITY COST ESTIMATION

Civil Engineering Portfolio Report

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Report Type	Quantity Take-Off & Cost Estimation
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TOTAL PROJECT ESTIMATED COST

Rs 20,685,492

(PKR Twenty Million, Six Hundred Eighty-Five Thousand)

1. Project Overview

This Quantity Cost Estimation (CEP) report documents the complete material take-off and cost analysis for a multi-storey reinforced concrete building. The estimation covers all major structural elements from substructure to superstructure, including footings, columns, plinth beams, floor beams, roof beams, floor slabs, roof slab, and stairs.

The quantities were computed using standard engineering formulas and bar bending schedules (BBS) for reinforcement, with concrete volumes based on element geometry. A 20% wastage factor was applied to all reinforcement quantities before cost estimation. Market rates current at the time of estimation were applied for all materials.

Structural Elements Covered

#	Structural Element	Element Type	Rebar Grades Used
1	Footings	Isolated / Pad Footings	#4, #5
2	Columns	RC Rectangular Columns	#5, #6, #8, ø3 Ties
3	Plinth Beams	Ground-Level Tie Beams	#4, #5, #6, ø3 Stirrups
4	Beams — Floors 1 & 2	RC Frame Beams	#4, #5, #6, ø3 Stirrups
5	Roof Beams	RC Roof-Level Beams	#4, #5, #6, ø3 Stirrups
6	Slabs — Floors 1 & 2	One-Way RC Slabs	#4, ø3
7	Roof Slab	One-Way RC Slab	#4, ø3
8	Stairs	RC Staircase	#4, ø3

2. Footings

Ten isolated pad footings of varying plan sizes and depths were quantified. Reinforcement was scheduled in both X and Y directions using #4 and #5 deformed bars as required by footing size.

2.1 Footing Schedule

ID	A (ft)	B (ft)	Depth	Rebar X	Bars X	Rebar Y	Bars Y	Conc. (cft)
F1	5'	6'	1.25'	#4 @ 10"	11	#4 @ 12"	13	64.69
F2	6'	7'	1.25'	#4 @ 14.4"	15	#4 @ 16.8"	18	87.13
F3	6.5'	7.5'	1.5'	#4 @ 19.5"	21	#4 @ 22.5"	24	99.94
F4	7'	8'	1.5'	#4 @ 21"	22	#4 @ 24"	25	58.66
F5	4.75'	8'	1.25'	#4 @ 14.25"	15	#4 @ 24"	25	146.19
F6	8'	9'	1.75'	#4 @ 24"	25	#4 @ 27"	28	25.06
F7	4'	4'	1.25'	#4 @ 12"	13	#4 @ 12"	13	142.16
F8	8'	8.75'	1.75'	#4 @ 24"	25	#4 @ 26.25"	27	200.56
F9	9'	11'	1.75'	#5 @ 21.6"	23	#5 @ 26.4"	27	214.09
F10	6.6'	16'	1.75'	#5 @ 13.2"	14	#5 @ 32"	33	1084.90

2.2 Footing Reinforcement Totals

Bar Size	Direction	Total Length (ft)
#4	X-Direction	1,639.34
#4	Y-Direction	1,618.94
#5	X-Direction	475.80
#5	Y-Direction	464.40
Total #4	Both	3,258.28 ft
Total #5	Both	940.20 ft

3. Columns

Seven column types were designed with varying bar arrangements, dimensions, and storey heights. Longitudinal bars (#5, #6, #8) and lateral ties (ø3 wire) were quantified for all columns.

3.1 Column Reinforcement Summary

Bar Type	#5 (ft)	#6 (ft)	#8 (ft)	ø3 Ties (ft)	Total Columns
All Column Types	677	11,730	1,544	6,840	27 columns

Parameter	Value
Concrete Volume	2,208 cft
Column Height (typical)	48 ft (4 floors)
Dowel Splice Length (ø5)	2.5 ft
Dowel Splice Length (ø6)	3.0 ft
Tie Spacing	5" — 6" typical

4. Beams

Beams were categorised into three groups: Plinth Beams (ground level), Floor Beams (Floors 1 & 2 combined), and Roof Beams. A total of 34 beam spans were scheduled per level, with three cross-section beam types used.

4.1 Beam Reinforcement Summary by Level

Level	#4 (ft)	#5 (ft)	#6 (ft)	ø3 Stirrups (ft)	Concrete (cft)
Plinth Beams	607.68	912.61	1,928.19	239	2,092.16
Beams (Floors 1 & 2)	1,823.04	2,641.83	5,919.33	717	6,276.49
Roof Beams	607.68	912.61	1,928.19	239	2,092.16
TOTAL	3,038.40	4,467.05	9,775.71	1,195	10,460.81

Note: Splice lengths included per ACI code requirements (#4 = 2 ft, #5 = 2.5 ft, #6 = 3 ft). Stirrup lengths account for cover and hook allowances.

5. Slabs & Stairs

5.1 Slab Schedule

ID	L (ft)	W (ft)	t (in)	Qty	ø3 Bars	Concrete (cft)
S-1	19.7	18.0	7"	4	1,793	827.40
S-2	19.7	18.0	7"	4	1,810	827.40
S-3	16.0	10.83	7"	1	679	101.08
S-4	18.0	8.7	7"	2	849	182.70
S-5	18.0	8.7	7"	3	693	274.05
S-6	7.0	19.7	7"	1	334	80.44
S-7	28.1	13.83	7"	1	586	226.70
TO TA L (Flo ors)	—	—	—	16	6,744	2,519.77
Ro of Sla b	Same layout	×1 level	7"	16	—	2,519.77

5.2 Staircase

Parameter	Value
ø3 Bars	891 ft
#4 Bars	930.8 ft
Concrete Volume	460.78 cft
Number of Flights	5

6. Material Quantity Summary

6.1 Reinforcement — Consolidated (with 20% Wastage)

Bar Type	Net Qty (ft)	With Wastage (ft)	Weight (Ton)	Unit Rate (Rs/Ton)
ø3 Ties	86,206	103,447	17.64	240,000
#4 Bars	18,030	21,637	6.56	240,000
#5 Bars	6,084	7,301	3.45	240,000
#6 Bars	21,506	25,807	17.58	240,000
#8 Bars	1,544	1,853	2.24	240,000

6.2 Concrete Volumes by Element

Structural Element	Volume (cft)	Volume (m³)
Footings	2,109	59.72
Columns	2,208	62.52
Plinth Beams	2,092.16	59.24
Beams (Floors 1&2)	6,276.49	177.73
Roof Beams	2,092.16	59.24
Slabs (Floors)	7,559.31	214.06
Roof Slab	2,519.77	71.35
Stairs	460.78	13.05
GRAND TOTAL	24,031.27	680.53

7. Cost Estimate

The following cost estimate is based on prevailing market rates for construction materials in Pakistan. All reinforcement quantities include a 20% wastage allowance. Concrete quantities account for material mix ratios per design specifications.

#	Material	Quantity	Unit Rate	Total Cost (Rs)
1	ø3 Tie Bars	17.64 Ton	Rs 240,000/Ton	Rs 4,234,315
2	#4 Rebar	6.56 Ton	Rs 240,000/Ton	Rs 1,573,409
3	#5 Rebar	3.45 Ton	Rs 240,000/Ton	Rs 828,990
4	#6 Rebar	17.58 Ton	Rs 240,000/Ton	Rs 4,219,701
5	#8 Rebar	2.24 Ton	Rs 240,000/Ton	Rs 538,538
6	Cement	4,794 Bags	Rs 1,500/Bag	Rs 7,191,000
7	Sand	332.9 Cft	Rs 200/Cft	Rs 66,581
8	Aggregates	665.8 Cft	Rs 200/Cft	Rs 133,162
9	Water	31,663 Liters	Rs 60/L	Rs 1,899,796
GRAND TOTAL				Rs 20,685,492

7.1 Cost Distribution by Category

Category	Cost (Rs)	% of Total
Reinforcement (all grades)	11,394,954	55.1%
Cement	7,191,000	34.8%
Water	1,899,796	9.2%
Sand & Aggregates	199,742	0.9%
TOTAL	20,685,492	100%

8. Methodology & Assumptions

8.1 Reinforcement Estimation

Bar Bending Schedules (BBS) were prepared for each structural element. Bar lengths were computed using element geometry plus standard cover allowances and hook lengths per ACI 318-14 requirements. Splice lengths were applied at construction joints. The following formulas were used:

Number of bars (X-dir) = (Footing width — 2×cover) / spacing + 1

Bar length = element span + 2 × (depth — cover)

Tie length = 2 × (X + Y dimensions) + hook allowances

8.2 Concrete Estimation

Concrete volumes were computed from gross element dimensions. No deductions were made for rebar volume (< 2%). Slab thicknesses are taken as 7 inches (0.583 ft) throughout.

8.3 Wastage & Cost Factors

Parameter	Value
Reinforcement wastage factor	20%
Concrete wastage factor	Not applied (included in mix)
Rebar unit rate (all grades)	Rs 240,000 / Ton
Cement rate	Rs 1,500 / Bag (50 kg)
Sand rate	Rs 200 / Cft
Aggregate rate	Rs 200 / Cft
Water rate	Rs 60 / Liter

— End of Report —

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